

```
'iptables -A OUTPUT -m state --state  
NEW,RELATED,ESTABLISHED -j ACCEPT'
```

```
'iptables -A INPUT -m state --state RELAT-  
ED,ESTABLISHED -j ACCEPT'
```

These open up the machine a little—allowing all outbound packets to get out. At this point, your machine will still not accept any other requests, such as those for ssh remote access. If you want to allow sshd to accept incoming connections on port 22, type in

```
'iptables -A INPUT -p tcp -m state --state  
NEW,ESTABLISHED -j ACCEPT -dport 22'
```

Similarly, if you're running the Squid proxy caching server on port 3128, you

could use the following command to allow access to other machines:

```
'iptables -A INPUT -p tcp -m state --state  
NEW,ESTABLISHED -j ACCEPT -dport 3128'
```

Of course, this is if you're using the default configurations of the services. For example, if you need to run Squid on port 8888, you'll need to change Squid's config file, and also set your firewall to accept connections on port 8888 by typing in `'iptables -A INPUT -p tcp -m state --state NEW,ESTABLISHED -j ACCEPT -dport 8888.'`

Remember, custom iptables commands work only as long as the computer is running; if you reboot it, you'll need

to re-enter all these. Or, you can save the commands as a script. iptables picks up the initial script from the `/etc/sysconfig/iptables` file. Once you're done setting up your firewall, backup this file and then use the `'iptables-save > /etc/sysconfig/iptables'` command to save your firewall settings.

If you don't fancy command-line interfaces, there are several GUI tools to use along with iptables. One of them is Firestarter (<http://firestarter.sourceforge.net>), which provides a point-and-click interface. Be warned, though—on Red Hat Linux 9, Firestarter isn't very stable.

The Toolbox

It's a little more fun securing a Linux-based computer than a Windows PC; there's just so many more good tools available—made by hackers of all sorts. Here are some powerful, essential tools that you should have. Read the man and info pages for all of them, to access detailed settings.

Chkrootkit (<http://www.chkrootkit.org/>): Rootkit is the term used for special tools by hackers, once they've gained access to a machine. A rootkit could be something like a tainted 'ps' program, which will never show a trojan or a backdoor; it

could be any binary written to masquerade as a legitimate Unix tool, but it is malicious. Think of a 'ps' program that has 'rm -rf *' written into it! To scan for such well-known rootkits and backdoors, use chkrootkit. Once installed, run the program and check whether it detects any rootkits. The sad thing is, if there are rootkits installed, there's no guarantee that all will be found; your only option is to remove the ones detected, check your system logs, and as a worst-case-scenario, clean format and reinstall Linux.

Nmap (<http://www.insecure.org/>) is a tool that maps your computer's ports as seen by others. It has numerous features, and can be tweaked to show a lot of detail. It's available on the Red Hat 9 CDs, but you should download and install the latest version. Use Nmap aggressively from a remote computer to check for the presence of offending trojans and backdoors. Combine it with 'netstat' and 'ps' to take out all such offending processes. We detected trojans on our test rig using Nmap.

F-Prot (<http://www.f-prot.com/>): Don't kid yourself into believing that your Linux machine doesn't need an anti-virus program. Worms and viruses come for all OSes. Get F-Prot, install it, and run it regularly. It comes free for use by home users, and is a command line scanning tool. Be sure to update it as often as you would for a Windows PC.

Tripwire (<http://www.tripwire.org/>) is a tool that checks file integrity. It checks signatures and several binary attributes for security. If the essential Linux program binaries have been attacked, tell-tale signs such as a change in the signature of the programs are seen. This is the sort of change that tripwire will detect.

PortSentry (<http://sourceforge.net/projects/sentrytools/>) is a tool that runs as a daemon and monitors TCP and UDP ports. PortSentry will block scanning hosts from making connections to your machine.

